Notes in sampling and Random-Walk algorithms

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Abstract

I. Introduction

Right now, this is only a small sketch of the things I will be doing in my stage de recherche. There will be errors (I hope that most only in language and not in mathematics or programming logic). The main purpose of these notes will be to understand more deeply the things I'm doing, creating a way to be always in-day with the things I do. Other than that, if I want to write something in the subject, this will be an easy repertory of contents, with the bibliography I read and the tools I found/created.

II. Foundations

Definition II.1 (Affine hull)**.** *The affine hull of a set S is the set of all afiine combinations of elements of S, that is:*

$$\operatorname{aff}(S) = \left\{ \sum_{i=1}^{k} \lambda_i x_i \;\middle|\; x_i \in S,\ \sum_{i=1}^{k} \lambda_i = 1 \right\}.$$

Definition II.2 (Convex hull)**.** *The convex hull of a set S is the (unique) minimal convex set containing X, given by:*

$$\operatorname{conv}(S) = \left\{ \sum_{i=1}^{k} \lambda_i x_i \;\middle|\; x_i \in S,\ \lambda_i \geq 0,\ \sum_{i=1}^{k} \lambda_i = 1 \right\}.$$

Definition II.3 (Polytope)**.** *A subset $P \subset \mathbb{R}^d$ is called a polytope if it is the set of solutions to a finite system of linear inequalities and is bounded.*

Theorem II.1 (Minkowski-Weyl)**.** *For a subset $P \subset \mathbb{R}^d$, the following statemetns are equivalent:*

- *P is a polytope, i.e, for some real (finite) matrix A and real vector b, $P = \{x : Ax \leq b\}$ and P is bounded,*
- *There exist finitely many points $v_1, \ldots, v_k \in \mathbb{R}^d$ such that $P = conv\{v_1, \ldots, v_k\}$.*

Then, one can see that every polytope have two representations, being the first known as (halfspace) H-representation and the latter (vertex) V-representation.

There is a well-defined equivalence between H and V-representations and some methods commonly used are:

- Double description method (H-rep $\rightarrow$ V-rep): Given a polytope in halfspace representation, the double description method enumerates the vertices and extreme rays of PP P. The algorithm incrementally adds inequalities one at a time, maintaining throughout the set of extreme generators of the current polyhedron. When a new inequality is introduced, generators that satisfy it are retained, infeasible generators are discarded, and new generators are created at intersections along the boundary hyperplane. The complexity depends on both the number of inequalities and the number of extreme generators; in degenerate cases, the number of intermediate generators may grow exponentially.
- Quickhull (V-rep $\rightarrow$ H-rep): Given a finite point set , the Quickhull algorithm computes the convex hull and returns its halfspace representation. Starting from an initial simplex, the algorithm recursively expands the hull by selecting points lying outside the current hull and updating the visible facets. The resulting facets define inequalities, whose collection yields the H-representation.

Definition II.4 (Feasibility and interior point)**.** *A polytope P is feasible if it is non-empty. A point $x \in P$ is an interior point if there exists a real $\epsilon > 0$ such that the ball $B(x, \epsilon) \subset P$ We denote the interior of P by* $\operatorname{int}(P)$.

Definition II.5 (Full-dimenstionality)**.** *A polytope $P \subset \mathbb{R}^d$ is full-dimensional if its affine hull equals $\mathbb{R}^d$, or equivalently, if P has non-empty interior. A polytope that is not full-dimensional is called degenerate.*

Definition II.6 (Inradius and circumradius)**.** *TODO*

Definition II.7 (Diameter)**.** *TODO*

Proposition II.8 (Boundedness and inner-ball property of full-dimensional polytope)**.** *Let $P \subset \mathbb{R}^d$ be a full-dimensional compact polytope, and let $x_0 \in \operatorname{int}(P)$ be an interior point. Then there exist constants $r, R > 0$ such that*

$$r\mathbb{B} \subseteq P - x_0 \subseteq R\mathbb{B},$$

where $\mathbb{B} = \{x \in \mathbb{R}^d : \|x\|_2 \leq 1\}$ is the unit Euclidean ball.

For sampling purposes, a reliable interior point for full-dimensional polytopes is the centroid of the vertices,

$$x_0 = \frac{1}{k} \sum_{i=1}^{k} v_i,$$

where $v_1, \ldots, v_k$ are the vertices of P.

Definition II.9 (Chord through a point)**.** *TODO*

Definition II.10 (Isotropic position)**.** *A full-dimensional convex body $P \subset \mathbb{R}^d$ is in isotrpic position if its centroid is at the origin and its covariance is a scalar multiple of the identity,*

$$\int_K x \, dx = 0, \qquad \int_K xx^\top dx = L_K^2 \, I_n,$$

Definition II.11 (Markov chain)**.** *A Markov chain on a state space Ω is a sequence of random variables $(X_0, X_1, \ldots)$ satisfying the Markov property: for all $t \geq 1$ and all measurable sets $A \subset \Omega$,*

$$\mathbb{P}(X_{t+1} \in A \mid X_0, \ldots, X_t) = \mathbb{P}(X_{t+1} \in A \mid X_t).$$

The future state depends only on the current state, not on the history.

Definition II.12 (Transition kernel)**.** *A transition kernel on a measurable space $(K, \mathcal{F})$ is a function*

$$P : K \times \mathcal{F} \to [0, 1]$$

such that:

1) *for every fixed $x \in K$, the map*

$$A \mapsto P(x, A)$$

   is a probability measure on $(K, \mathcal{F})$;

2) *for every fixed measurable set $A \in \mathcal{F}$, the map*

$$x \mapsto P(x, A)$$

   is measurable.

The quantity

$$P(x, A) = \mathbb{P}(X_{t+1} \in A \mid X_t = x)$$

represents the probability that the next state lies in A given that the current state is x.

Definition II.13 (Markov chain on a polytope)**.** *A Markov chain on a polytope $P \subset \mathbb{R}^d$ is a Markov chain with state space $\Omega = P$ and a transition kernel K that maps each interior point $x \in \operatorname{int}(P)$ to a probability mesure supported on P.*

Definition II.14 (Reference measure and target)**.** *TODO*

Definition II.15 (Stationary distribution)**.** *A probability measure π on Ω is stationary distribution of the Markov chain with transition kernel K if:*

$$\pi(A) = \int_\Omega K(x, A) \, \pi(dx) \qquad \text{for all measurable } A \subseteq \Omega.$$

Equivalently, $\pi K = \pi$: if $X_0 \sim \pi$, then $X_t \sim \pi$ for all $t \geq 0$.

Definition II.16 (Reversibility (detailed balance))**.** *TODO*

Proposition II.17 (Reversibility implies stationary)**.** *TODO*

Definition II.18 (Irreducibility)**.** *A Markov chain with transition kernel K and stationary distribution π is π-irreducible if for every state $x \in \Omega$ and every measurable set A with $\pi(A) > 0$, there exists $t \geq 1$ such that $K^t(x, A) > 0$. Informally, the chain can reach any region of positive measure from any starting point.*

Definition II.19 (Aperiodicity)**.** *A Markov chain is aperiodic if it does not cycle deterministic between disjoint subsets of* Ω*. Formally, the chain has period* 1 *at every state. Apariodicity, together with irretudcitbility, ensure convergence of* $K^t(x, \cdot)$ *to the stationary distribution as* $t \to \infty$*.*

Theorem II.2 (Ergodic theorem for Markov chain)**.** *TODO*

Definition II.20 (Total variation distance)**.** *TODO*

Definition II.21 (Mixing time)**.** *TODO*

Definition II.22 (Spectral gap)**.** *TODO*

Theorem II.3 (Spectral gap controls mixing)**.** *TODO*

Definition II.23 (Conductance (CHeeger coinstant))**.** *TODO*

Definition II.24 (Cheeger inequality)**.** *TODO*

Definition II.25 (Logconcave measure)**.**

With the built foundations, now we can address the main problem in this work: desgin a Markov chain in a polytope that is irreducible, aperiodic and mixes rapidly. To study this

III. Hit-and-Run: Algorithm, Theory and Convergence

IV. Sampling in Practice: How Many Points?

Definition IV.1 (Autocorrelation)**.**

Definition IV.2 (Effective sample size)**.**

Remark IV.3 (Thinning versus running longer)**.**

Definition IV.4 (Running mean distance to centroid)**.**

Definition IV.5 (Sample complexity)**.**

Remark IV.6 (Volume estimation as motivation)**.**

References

[1] TODO procurar dps